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Using (2) and substituting $\frac{2}{3}d$ for $m+p$ in (1) gives $\frac{5}{3}d = 2{,}500$, from which $d = 1{,}500$. Then from (1), $p+m = 1{,}000$, but $m = p - 100$ from (2), so $2p - 100 = 1{,}000$, $2p = 1{,}100$, and $p = 550$.

The correct answer is C.

$$3r \leq 4s + 5$$
$$|s| \leq 5$$

•

Given the inequalities above, which of the following CANNOT be the value of r?

A. −20

B. −5

C. 0

D. 5

E. 20

Algebra Inequalities

Since $|s| \leq 5$, it follows that $-5 \leq s \leq 5$. Therefore, $-20 \leq 4s \leq 20$, and hence $-15 \leq 4s + 5 \leq 25$. Since $3r \leq 4s + 5$ (given) and $4s + 5 \leq 25$ (end of previous sentence), it follows that $3r \leq 25$. Among the answer choices, $3r \leq 25$ is false only for $r = 20$.

The correct answer is E.

•

If m is an even integer, v is an odd integer, and $m > v > 0$, which of the following represents the number of even integers less than m and greater than v?

A. $\frac{m-v}{2} - 1$

B. $\frac{m-v-1}{2}$

C. $\frac{m-v}{2}$

D. $m - v - 1$

E. $m - v$

Arithmetic Properties of Numbers

Since there is only one correct answer, one method of solving the problem is to choose values for m and v and determine which of the expressions gives the correct number for these values. For example, if $m = 6$ and $v = 1$, then there are 2 even integers less than 6 and greater than 1, namely the even integers 2 and 4. As the table below shows, $\frac{m-v-1}{2}$ is the only expression given that equals 2.

$$\begin{aligned} \frac{m-v}{2} - 1 &= 1.5 \\ \frac{m-v-1}{2} &= 2 \\ \frac{m-v}{2} &= 2.5 \\ m - v - 1 &= 4 \\ m - v &= 5 \end{aligned}$$

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$$m - v = 5$$

To solve this problem it is not necessary to show that $\frac{m-v-1}{2}$ always gives the correct number of even integers. However, one way this can be done is by the following method, first shown for a specific example and then shown in general. For the specific example, suppose $v = 15$ and $m = 144$. Then a list—call it the first list—of the even integers greater than v and less than m is 16, 18, 20, …, 140, 142. Now subtract 14 (chosen so that the second list will begin with 2) from each of the integers in the first list to form a second list, which has the same number of integers as the first list: 2, 4, 6, …, 128. Finally, divide each of the integers in the second list (all of which are even) by 2 to form a third list, which also has the same number of integers as the first list: 1, 2, 3, …, 64. Since the number of integers in the third list is 64, it follows that the number of integers in the first list is 64. For the general situation, the first list is the following list of even integers: $v+1, v+3, v+5, \ldots, m-4, m-2$. Now subtract the even integer $v-1$ from (i.e., add $-v+1$ to) each of the integers in the first list to obtain the second list: 2, 4, 6, …, $m-v-3, m-v-1$. (Note, for example, that $m-4-(v-1) = m-v-3$.) Finally, divide each of the integers (all of which are even) in the second list by 2 to obtain the third list: 1, 2, 3, …, $\frac{m-v-3}{2}, \frac{m-v-1}{2}$.

Since the number of integers in the third list is $\frac{m-v-1}{2}$, it follows that the number of integers in the first list is $\frac{m-v-1}{2}$.

The correct answer is B.

•

A positive integer is divisible by 9 if and only if the sum of its digits is divisible by 9. If n is a positive integer, for which of the following values of k is $25 \times 10^n + k \times 10^{2n}$ divisible by 9?

A. 9

B. 16

C. 23

D. 35

E. 47

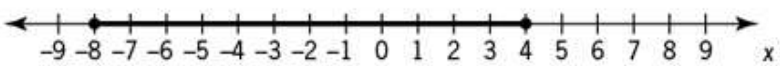
Arithmetic Properties of Numbers

Since n can be any positive integer, let $n = 2$. Then $25 \times 10^n = 2{,}500$, so its digits consist of the digits 2 and 5 followed by two digits of 0. Also, $k \times 10^{2n} = k \times 10{,}000$, so its digits consist of the digits of k followed by four digits of 0. Therefore, the digits of $(25 \times 10^n) + (k \times 10^{2n})$ consist of the digits of k followed by the digits 2 and 5, followed by two digits of 0. The table below shows this for $n = 2$ and $k = 35$:

$$\begin{aligned} 25 \times 10^n &= 2{,}500 \\ 35 \times 10^{2n} &= 350{,}000 \\ (25 \times 10^n) + (35 \times 10^{2n}) &= 352{,}500 \end{aligned}$$

Thus, when $n = 2$, the sum of the digits of $(25 \times 10^n) + (k \times 10^{2n})$ will be 2 + 5 = 7 plus the sum of the digits of k. Of the answer choices, this sum of digits is divisible by 9 only for $k = 47$, which gives 2 + 5 + 4 + 7 = 18. It can also be verified that, for each positive integer n, the only such answer choice is $k = 47$, although this additional verification is not necessary to obtain the correct answer.

The correct answer is E.



•

On the number line, the shaded interval is the graph of which of the following inequalities?

A. $|x| \leq 4$

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A. $|x| \leq 4$

B. $|x| \leq 8$

C. $|x-2| \leq 4$

D. $|x-2| \leq 6$

E. $|x+2| \leq 6$

Algebra Inequalities; Absolute Value

The midpoint of the interval from −8 to 4, inclusive, is $\frac{-8+4}{2} = -2$ and the length of the interval from −8 to 4, inclusive, is 4−(−8) = 12, so the interval consists of all numbers within a distance of $\frac{12}{2} = 6$ from −2. Using an inequality involving absolute values, this can be described by $|x-(-2)| \leq 6$, or $|x+2| \leq 6$.

Alternatively, the inequality $-8 \leq x \leq 4$ can be written as the conjunction $-8 \leq x$ and $x \leq 4$. Rewrite this conjunction so that the lower value, −8, and the upper value, 4, are shifted to values that have the same magnitude. This can be done by adding 2 to each side of each inequality, which gives $-6 \leq x+2$ and $x+2 \leq 6$. Thus, $x+2$ lies between −6 and 6, inclusive, and it follows that $|x+2| \leq 6$.

The correct answer is E.

•

Last year members of a certain professional organization for teachers consisted of teachers from 49 different school districts, with an average (arithmetic mean) of 9.8 schools per district. Last year the average number of teachers at these schools who were members of the organization was 22. Which of the following is closest to the total number of members of the organization last year?

A. 10^7

B. 10^6

C. 10^5

D. 10^4

E. 10^3

Arithmetic Statistics

There are 49 school districts and an average of 9.8 schools per district, so the number of schools is (49)(9.8) ≈ (50)(10) = 500. There are approximately 500 schools and an average of 22 teachers at each school, so the number of teachers is approximately (500)(22) ≈ (500)(20) = 10,000 = 10^4.

The correct answer is D.

•

Of all the students in a certain dormitory, $\frac{1}{2}$ are first-year students and the rest are second-year students. If $\frac{4}{5}$ of the first-year students have <u>not</u> declared a major and if the fraction of second-year students who have declared a major is 3 times the fraction of first-year students who have declared a major, what fraction of all the students in the dormitory are second-year students who have <u>not</u> declared a major?

A. $\frac{1}{15}$

B. $\frac{1}{5}$

C. $\frac{4}{15}$

D. $\frac{1}{2}$

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3

E. $\frac{2}{5}$

Arithmetic Applied Problems

Consider the table below in which T represents the total number of students in the dormitory. Since $\frac{1}{2}$ of the students are first-year students and the rest are second-year students, it follows that $\frac{1}{2}$ of the students are second-year students, and so the totals for the first-year and second-year columns are both $0.5T$. Since $\frac{4}{5}$ of the first-year students have not declared a major, it follows that the middle entry in the first-year column is $\frac{4}{5}(0.5T) = 0.4T$ and the first entry in the first-year column is $0.5T - 0.4T = 0.1T$. Since the fraction of second-year students who have declared a major is 3 times the fraction of first-year students who have declared a major, it follows that the first entry in the second-year column is $3(0.1T) = 0.3T$ and the second entry in the second-year column is $0.5T - 0.3T = 0.2T$. Thus, the fraction of students that are second-year students who have not declared a major is $\frac{0.2T}{T} = 0.2 = \frac{1}{5}$.

	First-year	Second-year	Total
Declared major	$0.1T$	$0.3T$	$0.4T$
Not declared major	$0.4T$	$0.2T$	$0.6T$
Total	$0.5T$	$0.5T$	T

The correct answer is B.

•

If the average (arithmetic mean) of x, y, and z is $7x$ and $x \neq 0$, what is the ratio of x to the sum of y and z?

A. 1:21

B. 1:20

C. 1:6

D. 6:1

E. 20:1

Algebra Ratio and Proportion

Given that the average of x, y, and z is $7x$, it follows that $\frac{x+y+z}{3} = 7x$, or $x + y + z = 21x$, or $y + z = 20x$. Dividing both sides of the last equation by $20(y+z)$ gives $\frac{1}{20} = \frac{x}{y+z}$, so the ratio of x to the sum of y and z is 1:20.

The correct answer is B.

•

Jonah drove the first half of a 100-mile trip in x hours and the second half in y hours. Which of the following is equal to Jonah's average speed, in miles per hour, for the entire trip?

A. $\frac{50}{x+y}$

100

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B. $\frac{100}{x+y}$

C. $\frac{25}{x}+\frac{25}{y}$

D. $\frac{50}{x}+\frac{50}{y}$

E. $\frac{100}{x}+\frac{100}{y}$

Algebra Applied Problems

Using average speed $=\frac{\text{total distance}}{\text{total time}}$, it follows that Jonah's average speed for his entire 100-mile trip is $\frac{100}{x+y}$.

The correct answer is B.

•

If the amount of federal estate tax due on an estate valued at \$1.35 million is \$437,000 plus 43 percent of the value of the estate in excess of \$1.25 million, then the federal tax due is approximately what percent of the value of the estate?

A. 30%

B. 35%

C. 40%

D. 45%

E. 50%

Arithmetic Percents; Estimation

The amount of tax divided by the value of the estate is

$$\frac{[0.437+(0.43)(1.35-1.25)]\ \cancel{\text{million}}}{1.35\ \cancel{\text{million}}} \quad \text{value as fraction}$$

$$=\frac{0.437+(0.43)(0.1)}{1.35} \quad \text{arithmetic}$$

$$=\frac{0.48}{1.35}=\frac{48}{135} \quad \text{arithmetic}$$

By long division, $\frac{48}{135}$ is approximately 35.6, so the closest answer choice is 35%.

Alternatively, $\frac{48}{135}$ can be estimated by $\frac{48}{136}=\frac{6}{17}\approx\frac{6}{18}=\frac{1}{3}\approx 33\%$, so the closest answer choice is 35%. Note that $\frac{48}{135}$ is greater than $\frac{48}{136}$, and $\frac{6}{17}$ is greater than $\frac{6}{18}$, so the correct value is greater than 33%, which rules out 30% being the closest.

The correct answer is B.

$$7x+6y\le 38{,}000$$
$$4x+5y\le 28{,}000$$

-

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•

A manufacturer wants to produce x balls and y boxes. Resource constraints require that x and y satisfy the inequalities shown. What is the maximum number of balls and boxes combined that can be produced given the resource constraints?

A. 5,000

B. 6,000

C. 7,000

D. 8,000

E. 10,000

Algebra Inequalities

We are to determine the maximum value of $x + y$ given the inequalities above. Note that if $A \leq B$ and $C \leq D$, then we can "add inequalities" to obtain $A + C \leq B + D$, since (roughly speaking) the sum of two smaller numbers is less than the sum of two larger numbers. Adding the inequalities shown above gives $(7x + 6y) + (4x + 5y) \leq 38{,}000 + 28{,}000$, or $11x + 11y \leq 66{,}000$. Dividing both sides of this last inequality by 11 gives $x + y \leq 6{,}000$. Therefore, the values of $x + y$ are at most 6,000, and hence the maximum value of $x + y$ is *at most* 6,000.

The fact that the maximum value of $x + y$ is *equal* to 6,000 follows from the fact that the system of simultaneous equations $7x + 6y = 38{,}000$ and $4x + 5y = 28{,}000$ has a solution, which in turn follows from the fact that these two equations correspond to a pair of nonparallel lines in the standard (x,y) coordinate plane. In particular, the pair $x = 2{,}000$ and $y = 4{,}000$ satisfy both the two inequalities and the equation $x + y = 6{,}000$.

The correct answer is B.

•

If $\frac{3}{10^4} = x\%$, then $x =$

A. 0.3

B. 0.03

C. 0.003

D. 0.0003

E. 0.00003

Arithmetic Percents

Given that $\frac{3}{10^4} = x\%$, and writing $x\%$ as $\frac{x}{100}$, it follows that $\frac{3}{10^4} = \frac{x}{100}$. Multiplying both sides by 100 gives $x = \frac{300}{10^4} = \frac{300}{10{,}000} = \frac{3}{100} = 0.03$.

The correct answer is B.

•

What is the remainder when 3^{24} is divided by 5?

A. 0

B. 1

C. 2

D. 3

E. 4

Arithmetic Properties of Numbers

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A pattern in the units digits of the numbers 3, $3^2 = 9$, $3^3 = 27$, $3^4 = 81$, $3^5 = 243$, etc., can be found by observing that the units digit of a product of two integers is the same as the units digit of the product of the units digit of the two integers. For example, the units digit of $3^5 = 3 \times 3^4 = 3 \times 81$ is 3 since the units digit of 3×1 is 3, and the units digit of $3^6 = 3 \times 3^5 = 3 \times 243$ is 9 since the units digit of 3×3 is 9. From this it follows that the units digit of the powers of 3 follow the pattern 3, 9, 7, 1, 3, 9, 7, 1, etc., with a units digit of 1 for 3^4, 3^8, 3^{12}, …, 3^{24}, …. Therefore, the units digit of 3^{24} is 1. Thus, 3^{24} is 1 more than a multiple of 10, and hence 3^{24} is 1 more than a multiple of 5, and so the remainder when 3^{24} is divided by 5 is 1.

The correct answer is B.

•

José has a collection of 100 coins, consisting of nickels, dimes, quarters, and half-dollars. If he has a total of 35 nickels and dimes, a total of 45 dimes and quarters, and a total of 50 nickels and quarters, how many half-dollars does he have?

A. 15

B. 20

C. 25

D. 30

E. 35

Algebra Simultaneous Equations

Letting n, d, q, and h, respectively, represent the numbers of nickels, dimes, quarters, and half-dollars José has, determine the value of h.

The following are given:

1. $n + d + q + h = 100$
2. $n + d = 35$
3. $d + q = 45$
4. $n + q = 50$

Adding (2), (3), and (4) gives $2n + 2d + 2q = 130$ or $n + d + q = 65$. Subtracting this equation from (1) gives $h = 100 - 65 = 35$.

The correct answer is E.

•

David used part of \$100,000 to purchase a house. Of the remaining portion, he invested $\frac{1}{3}$ of it at 4 percent simple annual interest and $\frac{2}{3}$ of it at 6 percent simple annual interest. If after a year the income from the two investments totaled \$320, what was the purchase price of the house?

A. \$96,000

B. \$94,000

C. \$88,000

D. \$75,000

E. \$40,000

Algebra Applied Problems; Percents

Let x be the amount, in dollars, that David used to purchase the house. Then David invested $(100{,}000 - x)$ dollars, $\frac{1}{3}$ at 4% simple annual interest and $\frac{2}{3}$ at 6% simple annual interest. After one year the total interest, in dollars, on this investment was $\frac{1}{3}(100{,}000 - x)(0.04) + \frac{2}{3}(100{,}000 - x)(0.06) = 320$. Solve this equation to find the

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this investment was $\frac{1}{3}(100{,}000 - x)(0.04) + \frac{2}{3}(100{,}000 - x)(0.06) = 320$. Solve this equation to find the value of x.

$$\begin{aligned}
\frac{1}{3}(100{,}000 - x)(0.04) + \frac{2}{3}(100{,}000 - x)(0.06) &= 320 && \text{given} \\
(100{,}000 - x)(0.04) + 2(100{,}000 - x)(0.06) &= 960 && \text{multiply both sides by 3} \\
4{,}000 - 0.04x + 12{,}000 - 0.12x &= 960 && \text{distributive property} \\
16{,}000 - 0.16x &= 960 && \text{combine like terms} \\
16{,}000 - 960 &= 0.16x && \text{add } 0.16x - 960 \text{ to both sides} \\
100{,}000 - 6{,}000 &= x && \text{divide both sides by 0.16} \\
94{,}000 &= x
\end{aligned}$$

Therefore, the purchase price of the house was $94,000.

The correct answer is B.

•

A certain manufacturer sells its product to stores in 113 different regions worldwide, with an average (arithmetic mean) of 181 stores per region. If last year these stores sold an average of 51,752 units of the manufacturer's product per store, which of the following is closest to the total number of units of the manufacturer's product sold worldwide last year?

A. 10^6

B. 10^7

C. 10^8

D. 10^9

E. 10^{10}

Arithmetic Estimation

$$\begin{aligned}
(113)(181)(51{,}752) &\approx (100)(200)(50{,}000) \\
&= 10^2 \times (2 \times 10^2) \times (5 \times 10^4) \\
&= (2 \times 5) \times 10^{2+2+4} \\
&= 10^1 \times 10^8 = 10^9
\end{aligned}$$

The correct answer is D.

•

Andrew started saving at the beginning of the year and had saved $240 by the end of the year. He continued to save and by the end of 2 years had saved a total of $540. Which of the following is closest to the percent increase in the amount Andrew saved during the second year compared to the amount he saved during the first year?

A. 11%

B. 25%

C. 44%

D. 56%

E. 125%

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Arithmetic Percents

Andrew saved \$240 in the first year and \$540 – \$240 = \$300 in the second year. The percent increase in the amount Andrew saved in the second year compared to the amount he saved in the first year is

$$\left(\frac{300-240}{240}\times 100\right)\% = \left(\frac{60}{240}\times 100\right)\% = \left(\frac{1}{4}\times 100\right)\% = 25\%.$$

The correct answer is B.

•

If x is a positive integer, r is the remainder when x is divided by 4, and R is the remainder when x is divided by 9, what is the greatest possible value of $r^2 + R$?

A. 25

B. 21

C. 17

D. 13

E. 11

Arithmetic Properties of Integers

If r is the remainder when the positive integer x is divided by 4, then $0 \le r < 4$, so the maximum value of r is 3. If R is the remainder when the positive integer x is divided by 9, then $0 \le R < 9$, so the maximum value of R is 8. Thus, the maximum value of $r^2 + R$ is $3^2 + 8 = 17$.

The correct answer is C.

•

Each of the nine digits 0, 1, 1, 4, 5, 6, 8, 8, and 9 is used once to form 3 three-digit integers. What is the greatest possible sum of the 3 integers?

A. 1,752

B. 2,616

C. 2,652

D. 2,775

E. 2,958

Arithmetic Place Value

To create 3 three-digit numbers using each of the digits 0, 1, 1, 4, 5, 6, 8, 8, and 9 and having the maximum possible sum, the greatest three digits must be in hundreds place, the next greatest three in tens place, and the three smallest digits in units place. The sum will then be (9 + 8 + 8)(100) + (4 + 5 + 6)(10) + (0 + 1 + 1) = 25(100) + 15(10) + 2 = 2,500 + 150 + 2 = 2,652.

The correct answer is C.

•

Given that $1^2 + 2^2 + 3^2 + \ldots + 10^2 = 385$, what is the value of $3^2 + 6^2 + 9^2 + \ldots + 30^2$?

A. 1,155

B. 1,540

C. 1,925

D. 2,310

E. 3,465

Arithmetic Series and Sequences

We can use the fact that each term of the second series is $3^2 = 9$ times greater than the corresponding term of the first series to find the sum of the second series.

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series to find the sum of the second series.

$3^2 + 6^2 + 9^2 + \ldots + 30^2$

$= (3 \cdot 1)^2 + (3 \cdot 2)^2 + (3 \cdot 3)^2 + \ldots + (3 \cdot 10)^2$

$= (3^2 \cdot 1^2) + (3^2 \cdot 2^2) + (3^2 \cdot 3^2) + \ldots + (3^2 \cdot 10^2)$

$= 3^2(1^2 + 2^2 + 3^2 + \ldots + 10^2) = 9(385) = 3{,}465$

The correct answer is E.

•

If water is leaking from a certain tank at a constant rate of 1,200 milliliters per hour, how many seconds does it take for 1 milliliter of water to leak from the tank?

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. 2

D. 3

E. 20

Arithmetic Measurement Conversion; Rate Problem

We are given that 1,200 milliliters leak in 1 hour, or 3,600 seconds. Because the leaking is at a constant rate, it takes $\frac{1}{1{,}200}$ as long for 1 milliliter to leak, and so 1 milliliter leaks in $\frac{3{,}600}{1{,}200} = 3$ seconds.

The correct answer is D.

•

When the positive integer k is divided by the positive integer n, the remainder is 11. If $\frac{k}{n} = 81.2$, what is the value of n?

A. 9

B. 20

C. 55

D. 70

E. 81

Arithmetic Properties of Integers

When the positive integer k is divided by the positive integer n, there exist unique positive integers q (the quotient) and r (the remainder) such that $k = qn + r$, where $0 \le r < n$. For example, when 43 is divided by 5, we have 43 = (8)(5) + 3 and $0 \le 3 < 5$. Dividing both sides of $k = qn + r$ by n and using $r = 11$ gives $\frac{k}{n} = q + \frac{r}{n}q + \frac{11}{n}$. We are given that $\frac{k}{n} = 81.2 = 81 + \frac{2}{10}$, so $q + \frac{11}{n} = 81 + \frac{2}{10}$. Therefore, $q = 81$ and $\frac{11}{n} = \frac{2}{10}$, or $n = 55$.

The correct answer is C.

•

The total area of a certain continent is approximately 3.8×10^{16} square inches. Which of the following is closest to the area of the continent, in square miles? (1 square mile is approximately 4.0×10^9 square inches.)

A. 6.7×10^5

B. 2.0×10^6

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B. 2.0×10

C. 9.5×10^6

D. 1.1×10^7

E. 9.5×10^8

Arithmetic **Measurement Conversion**

The following calculations make use of the given approximations.

$$4.0 \times 10^9 \text{ in}^2 \approx 1 \text{ mi}^2 \qquad \text{given}$$

$$1 \text{ in}^2 \approx \frac{1}{4.0 \times 10^9} \text{mi}^2 \qquad \text{divide both sides by } 4.0 \times 10^9$$

$$3.8 \times 10^{16} \text{ in}^2 \approx \frac{3.8 \times 10^{16}}{4.0 \times 10^9} \text{mi}^2 \qquad \text{multiply both sides by } 3.8 \times 10^{16}$$

$$3.8 \times 10^{16} \text{ in}^2 \approx \frac{3.8}{4.0} \times \frac{10^{16}}{10^9} \text{mi}^2 \qquad \text{use } \frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

$$3.8 \times 10^{16} \text{ in}^2 \approx \frac{19}{20} \times 10^{16-9} \text{ mi}^2 \qquad \text{use } \frac{a^b}{a^c} = a^{b-c}$$

$$3.8 \times 10^{16} \text{ in}^2 \approx 0.95 \times 10^7 \text{ mi}^2$$

$$3.8 \times 10^{16} \text{ in}^2 \approx 9.5 \times 10^6 \text{ mi}^2$$

The correct answer is C.

•

If $xy = 1$, what is the value of $\frac{2^{(x+y)^2}}{2^{(x-y)^2}}$?

A. 2

B. 4

C. 8

D. 16

E. 32

Algebra **Exponents; Simplifying Algebraic Expressions**

Using the property $\frac{2^a}{2^b} = 2^{a-b}$, $\frac{2^{(x+y)^2}}{2^{(x-y)^2}} = 2^{(x+y)^2-(x-y)^2}$. Also, since $(x+y)^2 - (x-y)^2 = (x^2 + 2xy + y^2) - (x^2 - 2xy + y^2) = 4xy$ and $xy = 1$, it follows that $\frac{2^{(x+y)^2}}{2^{(x-y)^2}} = 2^{4xy} = 2^{4\times 1} = 16$.

The correct answer is D.

•

Of the 20 members of a kitchen crew, 17 can use the meat-cutting machine, 18 can use the bread-slicing machine, and 15 can use both machines. If one member of the crew is to be chosen at random, what is the probability that the

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15 can use both machines. If one member of the crew is to be chosen at random, what is the probability that the member chosen will be someone who cannot use either machine?

A. 0

B. $\frac{1}{10}$

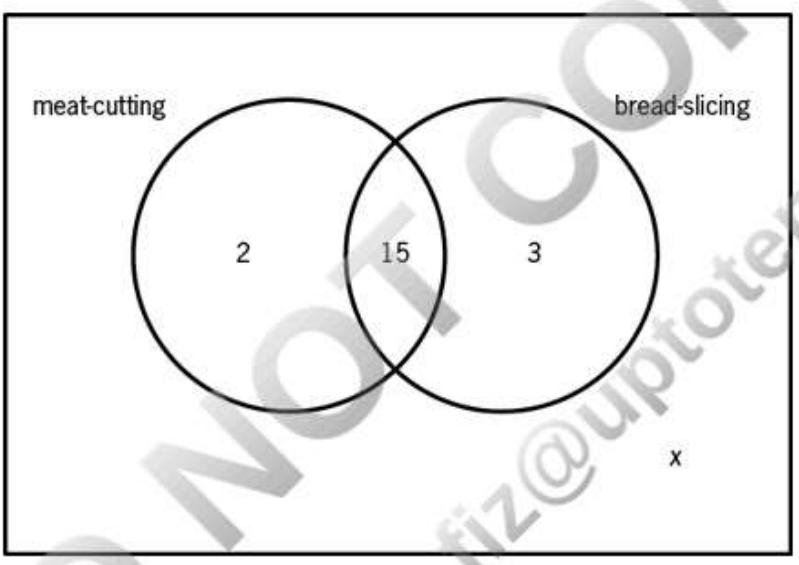
C. $\frac{1}{7}$

D. $\frac{1}{4}$

E. $\frac{1}{3}$

Arithmetic Probability; Sets (Venn diagrams)

Since 17 members can use the meat-cutting machine and 15 members can use both machines, 17 − 15 = 2 members can use only the meat-cutting machine. Similarly, since 18 members can use the bread-slicing machine and 15 members can use both machines, 18 − 15 = 3 members can use only the bread-slicing machine. The Venn diagram below shows these values, where x is the number of members who cannot use either machine.



Since there is a total of 20 members, it follows that $2 + 15 + 3 + x = 20$, or $x = 0$. Therefore, the probability that the member chosen cannot use either machine is $\frac{0}{20} = 0$.

The correct answer is A.

•

Which of the following is an integer?

I. $\frac{12!}{6!}$

II. $\frac{12!}{8!}$

III. $\frac{12!}{7!5!}$

A. I only

B. II only

C. III only

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D. I and II only

E. I, II, and III

Arithmetic Arithmetic Operations

I. $$\frac{12!}{6!} = \frac{\cancel{6!} \times 7 \times 8 \times 9 \times 10 \times 11 \times 12}{\cancel{6!}} = 7 \times 8 \times 9 \times 10 \times 11 \times 12,$$
which is an integer.

II. $$\frac{12!}{8!} = \frac{\cancel{8!} \times 9 \times 10 \times 11 \times 12}{\cancel{8!}} = 9 \times 10 \times 11 \times 12,$$
which is an integer.

III. $$\frac{12!}{7!5!} = \frac{\cancel{7!} \times 8 \times 9 \times 10 \times 11 \times \cancel{12}}{\cancel{7!} \times 1 \times 2 \times \cancel{3 \times 4} \times 5} = \frac{8 \times 9 \times \cancel{10} \times 11}{\cancel{2 \times 5}} = 8 \times 9 \times 11,$$
which is an integer.

Alternatively for III, note that $\frac{12!}{7!5!} = \binom{12}{7}$ is the number of combinations of 12 objects taken 7 at a time, and thus $\frac{12!}{7!5!}$ is an integer.

Therefore, the expression in each of I, II, and III is an integer.

The correct answer is E.

•

The function f is defined for all nonzero x by the equation $f(x) = x - \frac{1}{x}$. If $x \neq 0$, which of the following equals $f\left(\frac{1}{x}\right)$?

A. $f(x)$

B. $f(-x)$

C. $f\left(-\frac{1}{x}\right)$

D. $\frac{1}{f(x)}$

E. $-\frac{1}{f(x)}$

Algebra Functions

$$f\left(\frac{1}{x}\right) = \frac{1}{x} - \frac{1}{\frac{1}{x}} = \frac{1}{x} - x$$

$$f(-x) = -x - \frac{1}{-x} = -x + \frac{1}{x} = \frac{1}{x} - x$$

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Therefore, $f\left(\frac{1}{x}\right) = f(-x)$.

Alternatively, evaluate all of the expressions for an appropriate value of x, such as $x = 2$.

$$\begin{array}{rcrcrcr}
f\left(\frac{1}{x}\right) & = & f\left(\frac{1}{2}\right) & = & \frac{1}{2} - 2 & = & -\frac{3}{2} \\
f(x) & = & f(2) & = & 2 - \frac{1}{2} & = & \frac{3}{2} \\
f(-x) & = & f(-2) & = & -2 + \frac{1}{2} & = & -\frac{3}{2} \\
f\left(-\frac{1}{x}\right) & = & f\left(-\frac{1}{2}\right) & = & -\frac{1}{2} + 2 & = & \frac{3}{2} \\
\frac{1}{f(x)} & = & \frac{1}{f(2)} & = & \frac{1}{2 - \frac{1}{2}} & = & \frac{2}{3} \\
-\frac{1}{f(x)} & = & -\frac{1}{f(2)} & = & -\frac{1}{2 - \frac{1}{2}} & = & -\frac{2}{3}
\end{array}$$

The correct answer is B.

•

In the arithmetic sequence $t_1, t_2, t_3, \ldots, t_n, \cdots, t_1 = 23$ and $t_n = t_{n-1} - 3$ for each $n > 1$. What is the value of n when $t_n = -4$?

A. −1

B. 7

C. 10

D. 14

E. 20

Algebra Series and Sequences

From the given information, it follows that if we subtract 3 a total of $n - 1$ times from $t_1 = 23$, then we will obtain $t_n = -4$. Therefore, n satisfies the equation $23 - (n - 1)(3) = -4$. Now solve for n.

$$\begin{aligned}
23 - (n-1)(3) &= -4 && \text{from above} \\
23 - 3n + 3 &= -4 && \text{distributive property} \\
-3n &= -30 && \text{combine like terms} \\
n &= 10 && \text{divide both sides by } -3
\end{aligned}$$

Alternatively, subtract 3 from 23 until −4 is obtained and count the number of subtractions. One more than this number is the value of n such that $t_n = -4$.

The correct answer is C.

•

How many seconds will it take for a car that is traveling at a constant rate of 45 miles per hour to travel a distance of 220 yards? (1 mile = 1,760 yards)

A. 8

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B. 9

C. 10

D. 11

E. 12

Arithmetic Rate Problem; Measurement Conversion

First, convert 45 miles per hour to yards per second. We know that 1 hour is equal to 3,600 seconds giving us the fraction $\frac{45}{3{,}600}$. Divide both sides of the fraction by 3,600 to get the miles per second which equals $\frac{1}{80}$ miles in one second. We know that 1 mile is equal to 1,760 yards so the nest step is $1{,}760 \times \frac{1}{80}$. Therefore, it will take the car 1 second to travel 22 yards; hence it will take 10 seconds for the car to travel 220 yards.

Alternatively, let t be the number of seconds it will take for the car to travel 220 yards.

$$220 \text{ yd} = \left(45 \frac{\text{mi}}{\text{hr}}\right)(t \text{ sec}) \qquad \text{distance = rate} \times \text{time}$$

$$(220 \cancel{\text{yd}})\left(\frac{1 \text{ mi}}{1{,}760 \text{ yd}}\right) = \left(45 \frac{\text{mi}}{\cancel{\text{hr}}}\right)\left(\frac{1 \cancel{\text{hr}}}{3{,}600 \cancel{\text{sec}}}\right)(t \cancel{\text{sec}}) \qquad \begin{array}{l}1 \text{ mi} = 1{,}760 \text{ yd} \\ 1 \text{ hr} = 3{,}600 \text{ sec}\end{array}$$

$$\frac{220}{1{,}760} \text{ mi} = \left(\frac{45}{3{,}600} t\right) \text{mi} \qquad \text{cancel units}$$

$$\frac{2 \times 11 \times 10}{11 \times 16 \times 10} = \left(\frac{45}{45 \times 80}\right) t \qquad \text{factor}$$

$$\frac{1}{8} = \frac{t}{80} \qquad \text{cancel common factors}$$

$$t = 10 \qquad \text{multiply both sides by 80}$$

The correct answer is C.

•

A store's selling price of $2,240 for a certain computer would yield a profit of 40 percent of the store's cost for the computer. What selling price would yield a profit of 50 percent of the computer's cost?

A. $2,400

B. $2,464

C. $2,650

D. $2,732

E. $2,800

Algebra Applied Problems

Let $\$C$ be the store's cost for the computer. Then the store's profit from selling the computer would be $\$(2{,}240 - C)$, which is equal to 40 percent of the store's cost.

$$2{,}240 - C = 0.4C \qquad \text{given}$$

$$2{,}240 = 1.4C \qquad \text{add } C \text{ to both sides}$$

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$C = 1{,}600$ divide both sides by 1.4

Thus, the store's cost for the computer is \$1,600. Therefore, to yield a profit of 50 percent of the computer's cost, the selling price would have to be 1.5(\$1,600) = \$2,400.

The correct answer is A.

•

If a certain coin is flipped, the probability that the coin will land heads up is $\frac{1}{2}$. If the coin is flipped 5 times, what is the probability that it will land heads up on the first 3 flips and not on the last 2 flips?

A. $\frac{3}{5}$

B. $\frac{1}{2}$

C. $\frac{1}{5}$

D. $\frac{1}{8}$

E. $\frac{1}{32}$

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